

BookMyShow: Requirements & Operational Assumptions Matrix

Complete Architecture Specifications • Section A: Requirements Clarification Matrix

1. Functional Requirements Matrix

Capability Area	Functional Description	User Persona	Priority
Search & Browse	Search movies and theaters filtered by city, date, and showtime. View theater auditoriums and visual seat maps.	End Consumer	P0 (Critical)
Seat Selection & Hold	Real-time interactive seat selection with a 10-minute temporary lock during checkout to prevent duplicate claims.	End Consumer	P0 (Critical)
Bookings & Payments	Atomically process ticket payments via external gateways. Confirm seat status or trigger graceful release on failure.	End Consumer	P0 (Critical)
Cancellations & Refunds	Cancel confirmed bookings prior to movie cut-off time; release seats back to the public pool and trigger automated refund.	End Consumer	P1 (Core)
Digital Tickets & Alerts	Issue digital tickets with scannable QR codes; dispatch confirmation receipts and showtime reminders across Email, SMS, and Push.	End Consumer	P1 (Core)

2. Non-Functional Requirements (NFRs)

Quality Attribute	Target Metric / SLA	Architectural Strategy & Mitigation	Priority
Scalability	100,000 concurrent users	Stateless microservice auto-scaling on Kubernetes, CDN edge caching, in-memory Redis session state.	P0 (Critical)
Availability	99.9% Uptime target	Active-active multi-zone deployment, automated health checks, CDN-cached catalog browsing.	P0 (Critical)
Consistency	Zero double-bookings	Distributed temporary seat lock layer backed by relational ACID database transactions on final ticket commit.	P0 (Critical)
Latency	P95 < 300ms for booking APIs	Sub-80ms catalog searches via Redis/CDN caches; sub-200ms seat map retrieval; fast checkout finalization.	P0 (Critical)
Fault Tolerance	Zero financial data loss	Idempotent payment webhooks, circuit breakers on third-party gateways, and dead-letter queues (DLQ).	P0 (Critical)

3. Explicit System Assumptions

Auditorium & Seating Scale Average of 200 seats per screen, arranged across 3 pricing tiers (Silver, Gold, Recliner). An average theater contains 5 screens.	City-Level Data Volume Top metro cities average 50 theaters, 250 screens, and ~1,000 shows daily. Generates ~50 MB of new transactional records per city per day.	Checkout Hold Window Seats are temporarily reserved for exactly 10 minutes. If payment does not complete within this window, the hold expires automatically.
Read vs. Write Distribution Browsing showtimes and seat maps represents ~95% of traffic (~20:1 read-to-write ratio), heavily protected by cache tiers.	Cancellation Cut-Off Policy Cancellations are permitted up to 2 hours prior to movie start time, immediately returning seats to the available inventory pool.	External Gateway Resilience Payment gateways and notification vendors are assumed prone to intermittent downtime; all interactions enforce timeouts and retries.